

Weights or Skills?

A Survey of Robot-Learning Techniques—from Action-Predicting Weights to Robots that Write their Own Skills

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Robot learning is splitting into two bets: policies that bake competence into *frozen weights*—vision-language-action (VLA) models—and agents that write and refine their own *executable skills* as code. This survey organises the field around that axis, **weights or skills**, and makes its central contribution a deep-dive that arranges code-as-policy methods by their *degree of self-improvement*: from zero-shot program synthesis, through closed-loop self-repair and persistent skill memory, to the near-empty cell where execution feedback, skill memory, and evolutionary search combine into one open-ended loop. We anchor that frontier on ASPIRE and its real-world sibling ENPIRE, and we map the complementary “skills” pole—from unsupervised reinforcement-learning skill discovery to large-language-model skill libraries—showing that skills come in an RL flavour and a code flavour, of which only the code flavour self-improves without gradients. Finally, we connect the taxonomy to the emerging *skill economy*: commercial robot-skill marketplaces now distribute one-tap skills across robots, but ship only static playback, and we lay out the open problems—adaptation, cross-embodiment portability, provenance, safety verification, composition, and standardisation—that separate a store of animations from a store of capabilities. The survey covers roughly seventy-five systems across six technique families and provides a canonical taxonomy figure for each.

CCS Concepts: • **Computing methodologies** → **Robotic planning**; *Machine learning*.

Additional Key Words and Phrases: robot learning, code-as-policy, self-improvement, vision-language-action models, skill libraries, physical AI, survey

1 INTRODUCTION

Two paradigms now dominate robot learning. *Vision-language-action* (VLA) models predict actions from frozen weights; *code-as-policy* agents instead write executable programs and improve them from experience [32, 39]. This survey is organized around that fault line.

Why now: the skill stack. The app store for robot skills has arrived—Unitree UniStore ships one-tap, cross-model motion downloads [55]. But every skill in it is *canned playback*: the least-capable point on our map. The skill stack has three layers—**(1) Build** (how skills are created, §2–§3), **(2) Distribute** (marketplaces and hubs), and **(3) Adapt** (on-device self-improvement, §3.1). UniStore is layer 2 over static layer-1 skills; the promised value—“adjust grip per fruit so nothing bruises”—lives in layer 3, which no shipped skill does today.

Contributions. (i) a taxonomy of robot-learning techniques organized by *weights vs. skills* (§2); (ii) a deep-dive arranging code-as-policy by *degree of self-improvement* and isolating the un-surveyed §3.1.5 cell (§3.1); (iii) a companion map of the “skills” pole (§3.4); and (iv) an open-problems agenda for the emerging skill economy (§4).

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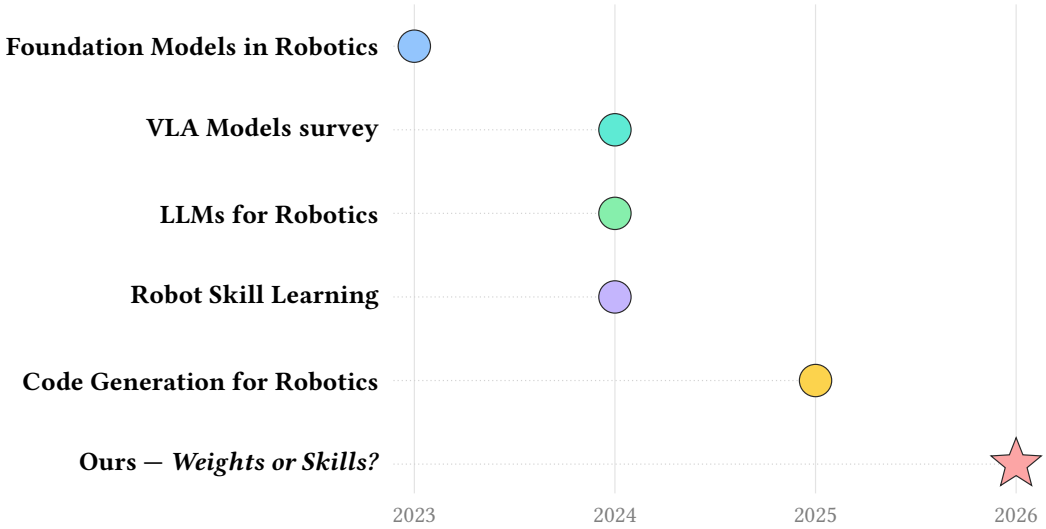


Table 1. Related surveys on a timeline. Prior surveys (2023–2025) each cover part of the landscape—foundation models, VLA policies, LLM planning, skill acquisition, code generation—but none organizes code-as-policy by *degree of self-improvement*. **Ours** (★, 2026) isolates the feedback + memory + search cell (§3.1.5) that ASPIRE occupies.

Related surveys. Table 1 places the closest prior surveys on a timeline. Each covers part of the landscape—foundation models, VLA policies, LLM-based planning, skill acquisition, or code generation—but none organises code-as-policy by *degree of self-improvement* or isolates the §3.1.5 cell, which is the gap this survey fills.

2 A TAXONOMY OF ROBOT-LEARNING TECHNIQUES

Figure 1 organizes the field into six branches. The dividing question is *what ships*: frozen weights (§3.2) or executable skills (§3.1). We tag mechanisms by **Feedback**, **Search**, and **Memory** throughout.

3 THE TECHNIQUE FAMILIES

We now examine each of the six branches of the taxonomy (Figure 1) in turn. We begin with the code-as-policy camp (§3.1)—the focus of this survey and the only family we organise by *degree of self-improvement*—and then survey the weights-based (§3.2), reward-synthesis (§3.3), skill-library (§3.4), transfer (§3.5), and benchmark (§3.6) families that surround it.

3.1 Code-as-Policy: Robots that Write their Own Skills

The organising contribution of this survey is to arrange code-as-policy methods not by application or robot embodiment, but by *how much of their own competence a system produces at run time*. Concretely, we ask five questions in sequence: does the agent (i) write control code at all, (ii) repair that code from execution feedback within a task, (iii) remember validated code across tasks, (iv) search a population of programs rather than following a single trajectory of repairs, and (v) close all of these into one open-ended loop? Each “yes” is a rung on the ladder of Figure 2, and the population thins sharply toward the top: of the thirteen zero-shot systems, only a handful ever reach the combined feedback–memory–search regime. The novelty of the survey is naming this progression and showing that the top rung is nearly empty.

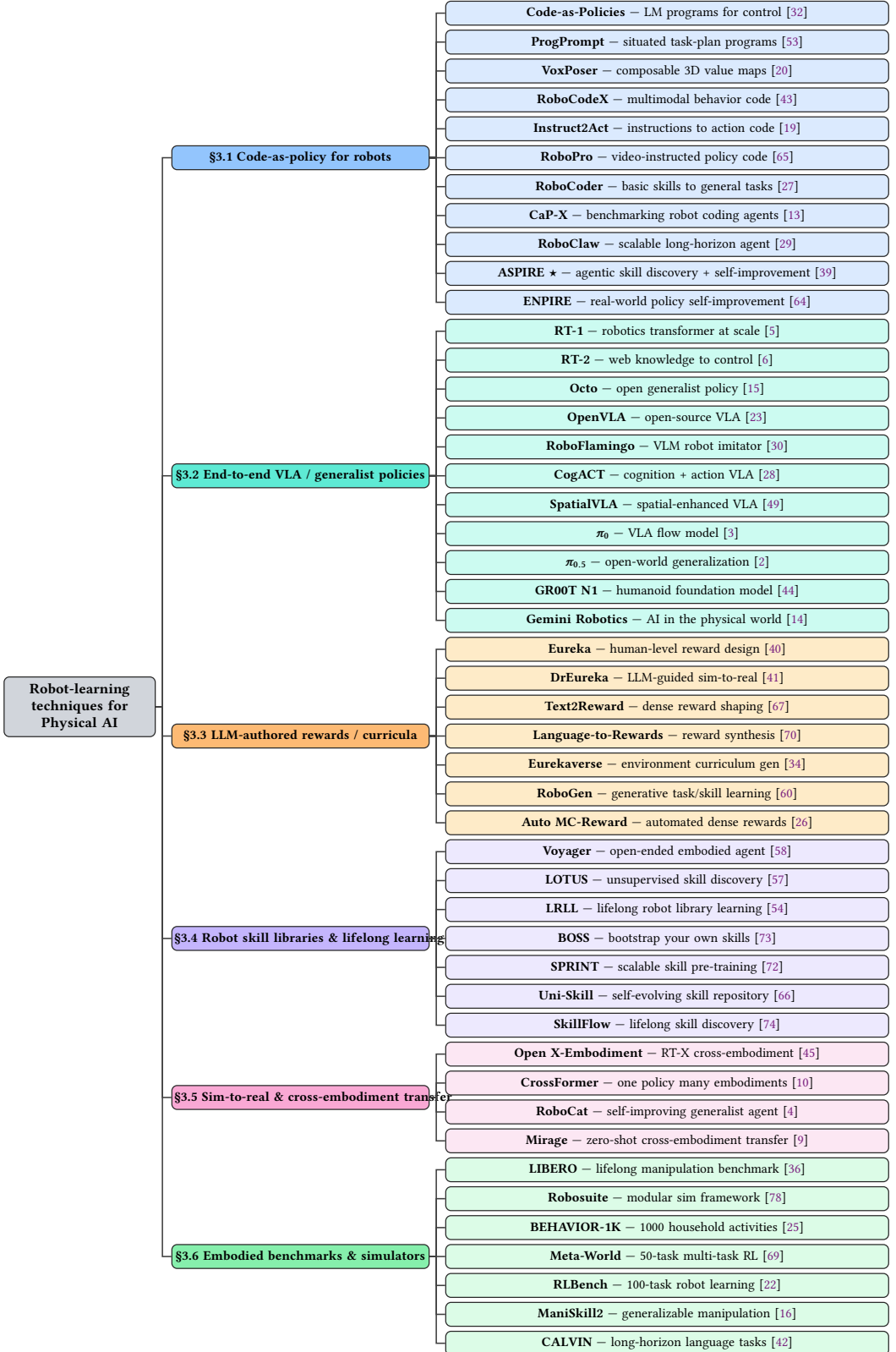


Fig. 1. Taxonomy of robot-learning techniques for Physical AI (~47 systems). The fault line: §3.1 ships *code* and §3.2 ships *weights*. **ASPIRE ★** anchors the self-improving code-as-policy frontier (§3.1).

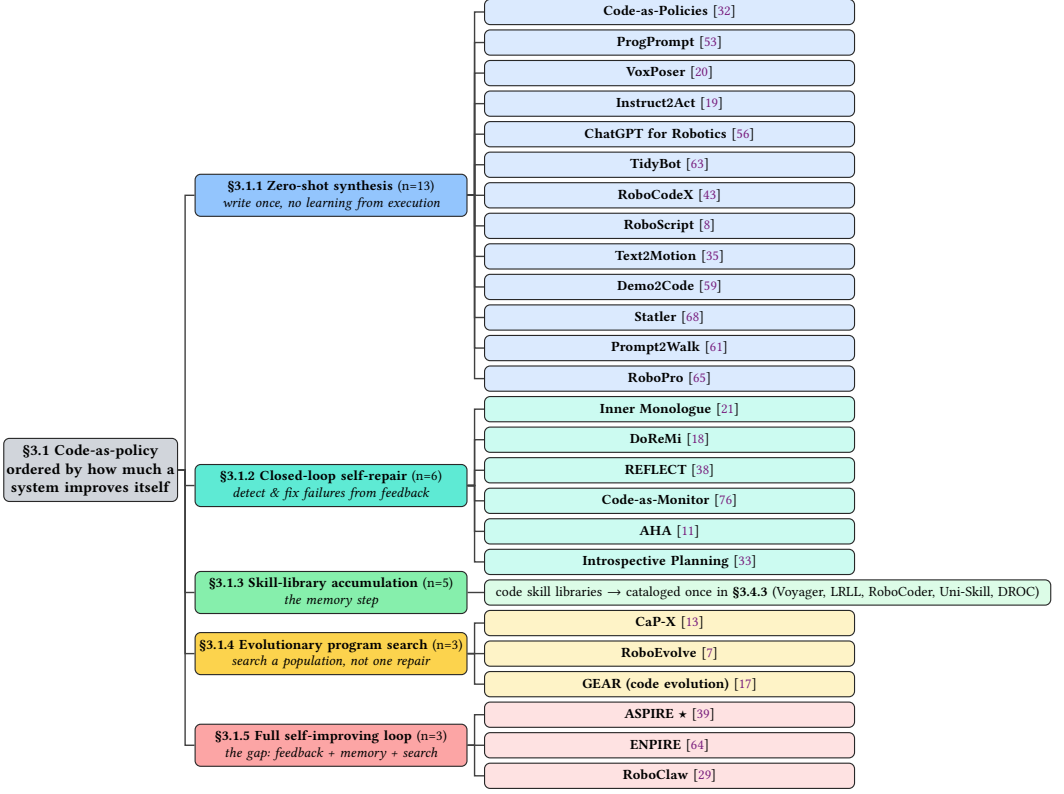


Fig. 2. The self-improvement funnel within code-as-policy ($13 \rightarrow 6 \rightarrow 5 \rightarrow 3 \rightarrow 3$), one colour per sub-section. As the demand rises (write once $\rightarrow$ repair $\rightarrow$ remember $\rightarrow$ search $\rightarrow$ all three), occupants thin out; §3.1.5—feedback + memory + search combined—is nearly empty, and only **ASPIRE ★** was built for it end-to-end.

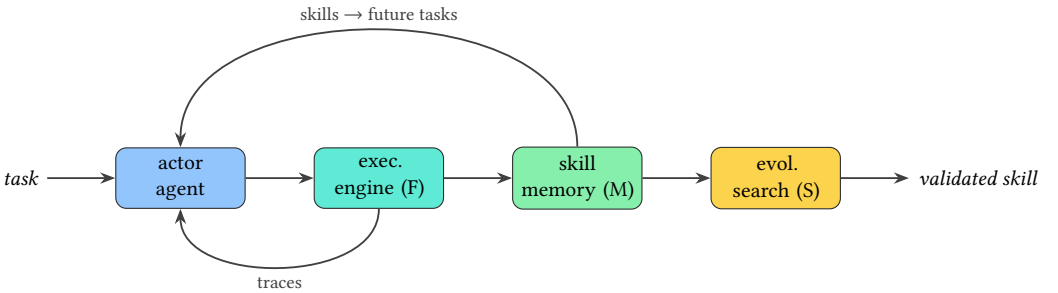


Fig. 3. The full self-improving loop (§3.1.5): feedback (F) + memory (M) + search (S) combine into one open-ended loop—the ASPIRE cell. VLAs (§3.2) have no such loop.

3.1.1 Zero-shot synthesis. The foundational move is to treat the language model as a *program synthesiser* over a fixed library of perception and control primitives. *Code-as-Policies* [32] showed that an LLM prompted with an API and a natural-language instruction can emit executable Python that composes those primitives, recursively defining undefined functions and parameterising

control with arithmetic and feedback logic. *ProgPrompt* [53] situates the generated program in the scene by exposing available actions and objects as importable symbols, while *VoxPoser* [20] has the model write code that *composes 3D value maps* for a motion planner rather than calling fixed skills, loosening the dependence on a hand-authored API. Subsequent work broadened the input modality and task horizon—*RoboCodeX* [43] conditions on multimodal observations, *Instruct2Act* [19] maps instructions to perception–action code, and video- and demonstration-conditioned variants such as *RoboPro* [65], *Demo2Code* [59] and *Text2Motion* [35] recover programs from richer context. *Statler* [68] maintains an explicit world state across steps, and *TidyBot* [63] and *ChatGPT for Robotics* [56] demonstrate personalisation and prompt design for real hardware. What unites this rung, and limits it, is that the program is written *once*: there is no channel from execution back to the synthesiser, so a plan that fails at run time simply fails.

3.1.2 Closed-loop self-repair. The second rung adds a within-task feedback loop. *Inner Monologue* [21] feeds success detectors, scene descriptions and human feedback back into the model as language, letting it replan when a step fails. *DoReMi* [18] detects misalignments between plan and execution and triggers recovery, and *REFLECT* [38] builds a hierarchical, multimodal summary of past interaction that an LLM queries to explain a failure and propose a correction. More recent systems push the monitor into the perception loop—*Code-as-Monitor* [76] compiles spatio-temporal constraints into vision-language checks, and *AHA* [11] trains a model to reason explicitly about failure modes. The novelty of this rung is the closed loop, but it is a *short* loop: corrections are discarded at task boundaries, so nothing accumulates.

3.1.3 Skill-library accumulation. The third rung makes the loop persistent by writing validated code into a growing library that later tasks retrieve and compose. Because this “memory” axis is a full technique family in its own right, we catalogue its systems once, in §3.4 (§3.4.3)—*Voyager* [58] is the canonical example, curating an ever-growing library of verified skill programs with when-to-apply guards, while related methods distil language corrections into retrievable knowledge for future tasks [71]. The key distinction from §3.1.2 is temporal scope: competence compounds *across* tasks rather than being rebuilt each time.

3.1.4 Evolutionary program search. The fourth rung replaces a single chain of repairs with population-based search over programs. Rather than iteratively debugging one candidate, the agent maintains several, scores them by execution, and mutates the best. *CaP-X* [13] provides an interactive gym and benchmark for exactly this style of coding agent, and *RoboEvolve* [7] couples a planner and a learned simulator into a co-evolutionary loop that mines near-miss failures to stabilise search. The novelty is the shift from *repair* to *search*: exploration is explicit and parallel, which trades compute for a better chance of escaping local failures.

3.1.5 The full self-improving loop. The top rung combines all three mechanisms—execution feedback (F), a persistent skill memory (M), and evolutionary search (S)—into one open-ended loop (Figure 3). *ASPIRE* [39] instantiates this with a robot execution engine that emits per-primitive multimodal traces, a skill library of validated code with guards, and a population search over candidate programs, so that experience compounds and transfers across tasks and embodiments. Its real-world sibling *ENPIRE* [64] closes the same loop on physical hardware, using automatic reset, parallel rollouts, and log-driven algorithm revision to drive a coding agent to near-perfect success on dexterous tasks. This cell—feedback *and* memory *and* search—is what no prior survey isolates, and it is nearly empty: it is the flag this survey plants.

3.2 End-to-End Vision-Language-Action Models

The dominant alternative to writing code is to regress actions directly from pixels and language with a single large network, so that competence lives in *frozen weights* rather than in an inspectable program. Architecturally these vision-language-action (VLA) models share a backbone–action-head decomposition: a vision-language backbone encodes the scene and instruction, and a pluggable action head maps that representation to motor commands [3, 23].

The lineage begins with *RT-1* [5], a transformer trained on large-scale real robot data that discretises continuous actions into categorical bins. *RT-2* [6] then co-fine-tunes an Internet-pretrained vision-language model on robot trajectories, emitting actions as text tokens and inheriting semantic generalisation from web data. Open reproductions followed: *Octo* [15] and *OpenVLA* [23]—the latter a 7B model with a DINOv2/SigLIP vision stack—are trained on the pooled Open X-Embodiment corpus (§3.5), and *RoboFlamingo* [30] adapts a vision-language model into an imitation policy.

A second design axis is the *action representation*. Discrete token heads (RT-1/RT-2) are simple but coarse; recent systems favour continuous heads. π_0 [3] attaches a *flow-matching* head to a PaLI-Gemma backbone for smooth, high-frequency bimanual control, and $\pi_{0.5}$ [2] extends it toward open-world generalisation; *CogACT* [28] separates a cognition module from a diffusion-transformer action module, and *SpatialVLA* [49] injects 3D spatial encodings. Flow- and diffusion-based heads trade a few extra parameters for far fewer inference steps and better high-degree-of-freedom precision.

At the largest scale, foundation-model efforts target whole platforms: *GR00T N1* [44] pairs a slow System-2 vision-language reasoner with a fast System-1 diffusion transformer that produces motor actions at high rate for humanoids, and *Gemini Robotics* [14] builds a VLA together with an embodied-reasoning model on the Gemini backbone. Across all of these, the defining property—and the reason we treat them as the *foil* for this survey—is that there is *no loop and no code*: competence is acquired by scaling data and parameters, generalises by interpolation rather than search, and cannot be edited, audited, or recombined after training.

3.3 Reward and Curriculum Synthesis

A third family keeps the language model in the role of programmer but points it at a different artefact: instead of policy code, it writes the *reward*, *environment*, or *curriculum* that a reinforcement-learning loop then compiles into a policy. A feedback loop exists—the model reads back training statistics and revises—but what ships is still weights, which is why we separate this family from the self-improving code camp (§3.1).

Eureka [40] is the canonical instance: given unmodified environment source code and a task description, a coding LLM zero-shot generates an executable reward function, then improves it by *evolutionary search* guided by *reward reflection*—a textual summary of training statistics—exceeding expert human rewards on 83% of a 29-task IsaacGym suite. *DrEureka* [41] extends the idea to sim-to-real by also synthesising domain-randomisation ranges, achieving zero-shot real-world locomotion. *Text2Reward* [67] generates dense reward programs grounded in a compact environment representation and refines them from human feedback, matching or beating expert-written rewards on most of a seventeen-task suite, while *Language to Rewards* [70] uses the reward function as the interface between an LLM and a MuJoCo model-predictive controller.

Beyond the reward itself, the same generative recipe is applied to the *training environment*. *Eurekaverse* [34] has an LLM propose a progressively harder curriculum of environments—learning parkour skills that transfer to a real robot—and *RoboGen* [60] runs a propose-generate-learn cycle that fabricates tasks, scenes, and supervision in simulation to unlock effectively unlimited training data; *Auto MC-Reward* [26] automates dense rewards for open-world Minecraft. The through-line

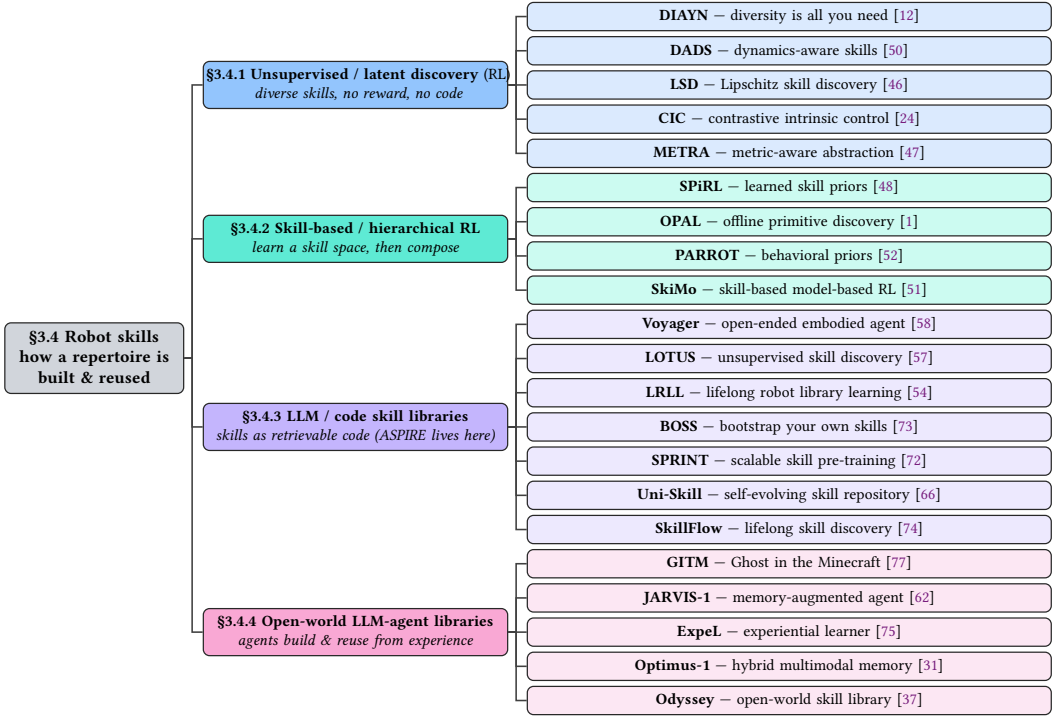


Fig. 4. The “skills” pole, one colour per sub-section. §3.4.1–2 build skills as *RL policies* (mature, included as landscape); §3.4.3–4 build them as *retrievable code*, where ASPIRE’s self-improving §3.1 lives.

is that language becomes a bridge from high-level intent to low-level control *via* an RL optimiser, rather than producing an executable, reusable skill directly.

3.4 Skill Libraries: How a Repertoire is Built

Where §3.1 asked *how* a code skill improves, this family asks the prior question of *how a repertoire of skills is built at all*. Figure 4 organises the answer along one axis—skills learned as latent reinforcement-learning policies versus skills stored as retrievable code—and only the code flavour self-improves without gradients, which links this family back to §3.1.

3.4.1 Unsupervised and latent skill discovery. The oldest lineage learns a diverse set of skills with *no task reward and no code*, by maximising an information-theoretic objective. *DIAYN* [12] maximises the mutual information between a latent skill code and the visited states, yielding distinguishable behaviours; *DADS* [50] makes the discovered skills dynamics-aware and hence usable for model-based control; and *LSD* [46], *CIC* [24], and *METRA* [47] push toward more dynamic, far-reaching, and scalable skills through Lipschitz constraints, contrastive objectives, and metric-aware abstractions respectively.

3.4.2 Skill-based and hierarchical reinforcement learning. A second lineage extracts a continuous *skill space* from offline data and then plans or learns within it. *SPiRL* [48] learns a skill embedding together with a skill prior that accelerates downstream RL; *OPAL* [1] discovers temporally extended primitives for offline RL; *PARROT* [52] learns an invertible behavioural prior; and *SkiMo* [51] learns a skill dynamics model so that planning happens directly in skill space.

3.4.3 LLM and code skill libraries. The code flavour—where the self-improving methods of §3.1 live—stores skills as retrievable programs. *Voyager* [58] curates an ever-growing library of verified code skills in Minecraft; *LOTUS* [57] performs continual imitation learning through unsupervised skill discovery; *LRL* [54] bootstraps a composable robot library with language models; and *BOSS* [73] and *SPRINT* [72] grow skill repertoires by LLM-guided bootstrapping and instruction relabelling. The most recent entries close the loop between video and code: *Uni-Skill* [66] builds a self-evolving skill repository from unstructured robot video, and *SkillFlow* [74] benchmarks lifelong discovery, patching, and reuse of a skill library.

3.4.4 Open-world LLM-agent libraries. Finally, embodied LLM agents accumulate skills from open-ended experience. *GITM* [77] and *JARVIS-1* [62] pair a planner with text or multimodal memory in Minecraft; *ExpeL* [75] extracts reusable insights from a stream of trials; *Optimus-1* [31] adds a hybrid multimodal memory for long-horizon tasks; and *Odyssey* [37] equips an agent with an open-world library of primitive and compositional skills. These systems foreshadow the marketplace dynamics of §4: a library that grows with use.

3.5 Sim-to-Real and Cross-Embodiment Transfer

A skill is only useful if it moves—across the simulation-to-reality gap and across robot bodies. The enabling artefact is scale: *Open X-Embodiment* [45] pools sixty datasets from twenty-one institutions into over a million trajectories spanning twenty-two embodiments and hundreds of skills, and shows that RT-X models trained on the mixture exhibit positive transfer and emergent capabilities across platforms.

Given such data, a single network can control very different bodies. *CrossFormer* [10] trains one transformer on 900K trajectories across twenty embodiments—arms, wheeled robots, quadrotors, and quadrupeds—without manually aligning observation or action spaces, and matches specialist policies tailored to each robot. Other work attacks transfer at test time rather than through data: *Mirage* [9] achieves *zero-shot* cross-embodiment transfer by “cross-painting”—masking the target robot and inpainting the source robot at the same pose so the policy sees a familiar body—and bridging the control gap with forward dynamics.

The most suggestive point for this survey is *RoboCat* [4], a goal-conditioned multi-embodiment agent that adapts to a new task from as few as a hundred demonstrations and then *generates its own data* for the next training round, forming a rudimentary self-improvement loop of the kind §3.1 makes explicit. This is also the marketplace question of §4 in disguise: a skill advertised for the G1, H1, B2, and Go2 platforms must survive exactly the cross-embodiment gap these methods study.

3.6 Benchmarks and Evaluation

Progress in the families above is measured on a shared set of simulated benchmarks, most of which report a single-number success rate. *LIBERO* [36] targets lifelong manipulation and knowledge transfer across task suites; *Meta-World* [69] offers fifty manipulation tasks for multi-task and meta-reinforcement learning; *RLBench* [22] provides a hundred vision-based tasks with demonstrations; and *ManiSkill2* [16] adds GPU-parallel simulation over twenty task families. *Robosuite* [78] is the modular MuJoCo framework underlying many of these, and *CALVIN* [42] evaluates long-horizon, language-conditioned control by the average number of sub-tasks completed in a chain. At the high-realism end, *BEHAVIOR-1K* [25] specifies a thousand everyday household activities in a photorealistic simulator, scoring both success and efficiency.

Table 2 contrasts one exemplar per camp on the axes that separate them. The recurring gap—and a direct motivation for the open problems of §4—is that these suites measure *one-shot competence*: they report whether a policy succeeds, not whether it *improves with experience*. No standard benchmark

Dimension	ASPIRE [39] <i>code+skills</i>	π_0 [3] <i>VLA weights</i>	Code-as-Pol. [32] <i>code, no mem.</i>	Eureka [40] <i>reward</i>
What it outputs	skill code+lib	action chunks	policy code	reward code
How it improves	F+M+S loop	gradient pretrain	– one-shot	RL+reflection
Gradient training?	✗	✓	✗	✓
Persistent skill library?	✓	✗	✗	✗
Failure feedback signal	exec. traces	–	–	train. stats
Search strategy	evolutionary	–	–	–
Real-robot validation	✓	✓	✓	✓
Continual / open-ended?	✓	✗	✗	✗
Beyond fixed APIs?	✓	✓	✗	–
Interpretable / editable?	✓	✗	✓	✓

Table 2. Capability matrix, one exemplar per camp; each method header links to its entry in the References. Only ASPIRE combines a persistent skill library with feedback- and search-driven self-improvement. ✓ = yes, ✗ = no, – = n/a.

yet plots held-out success as a function of accumulated interaction, which is exactly the quantity the self-improvement ladder of §3.1 is designed to raise.

4 THE SKILL ECONOMY: OPEN PROBLEMS

The arrival of a commercial marketplace for robot skills—Unitree’s UniStore ships one-tap, cross-model motion packages [55]—turns several research questions from hypothetical into pressing. We frame them as the open agenda of this survey.

The static-to-adaptive gap. Every skill shipped today is, in effect, replayed: a motion package or a fixed program is installed and executed without perceiving whether *this* kitchen, gripper, or object differs from the one it was authored for. The techniques of §3.1—within-task repair, persistent memory, and search—are precisely what would let a downloaded skill *adapt* on the target robot. Closing this gap is the difference between a store of animations and a store of capabilities, and it is the direct application of the §3.1.5 loop to a distribution setting where the base skill is given rather than discovered.

Cross-embodiment portability. UniStore advertises a single skill running across the G1 and H1 humanoids and the B2/Go2 quadrupeds—bodies with different morphologies, action spaces, and dynamics. This is the cross-embodiment transfer problem (§3.5) at deployment scale: without a shared action interface [45] or explicit embodiment adaptation, a package tuned for one platform has no guarantee of correctness on another. Formal portability—what must hold for a skill to be certified on a new body—remains open.

Provenance and trust. A marketplace invites third-party, crowdsourced uploads, including raw motion-capture data and code from unknown authors. Robot skills act on the physical world, so provenance (who authored a skill, from what data, validated how) becomes a safety property, not merely a metadata nicety. There is no accepted standard for signing, attesting, or auditing a robot skill’s origin.

Safety verification. Vendors promise that “every skill is scanned, tested, and verified,” but for a code-as-policy skill this is undecidable in general and hard in practice: the skill’s effect depends

on the scene it meets. Practical questions include which pre-conditions a skill must declare, what sandboxed or simulated screening it must pass, and how run-time monitors (cf. *Code-as-Monitor* [76]) can bound behaviour on unseen inputs.

Skill composition. Downloading two skills should ideally yield a third: “make coffee” plus “clear the table” composed into a morning routine. Composition requires shared interfaces and a calculus over pre-/post-conditions; today’s motion packages are opaque and do not compose, and code skills compose only within a single authoring system.

Portability standards and skill ontologies. Reuse at scale needs a shared vocabulary: declared pre-conditions, expected effects, and embodiment-capability profiles, with relations such as *requires*, *provides*, and *substitutable-by*. Prior work on skill ontologies and hardware-level reusability points the way, but no cross-vendor standard exists, which is what a genuine ecosystem would require.

Local versus shared libraries. Finally, the field must reconcile two notions of “skill library” that this survey has kept distinct: the *self-built* library a robot grows from its own experience (ASPIRE, §3.1) and the *shared* library a robot downloads from a marketplace (UniStore). The interesting regime is the hybrid—an agent that both contributes to and draws from a commons—and its incentives, quality control, and feedback dynamics are entirely unstudied.

5 LIMITATIONS OF THIS SURVEY

We state the boundaries of the survey explicitly. First, our organising axis—*degree of self-improvement*—is deliberately code-centric; it foregrounds systems that emit and revise programs and treats weight-based policies (§3.2) and reinforcement-learning skill discovery (§3.4) as context rather than as the object of the taxonomy. A survey centred on representation learning or on real-world data collection would draw the map differently. Second, the frontier we emphasise (§3.1.5) is populated by very recent, in some cases concurrent, systems; benchmark numbers across them are not directly comparable, and we therefore report capabilities qualitatively rather than ranking methods. Third, the field moves quickly: several 2025–2026 systems we cite were released while this survey was being written, and the coverage should be read as a snapshot. Finally, the “skill economy” framing (§4) is grounded in an emerging commercial trend (one vendor’s marketplace at the time of writing); we treat it as motivation for open problems, not as a mature body of results.

6 FUTURE DIRECTIONS

Three directions follow directly from the map. **(1) Standardised evaluation of self-improvement.** The systems on the top rung (§3.1.5) each report improvement on their own splits; the field lacks a shared protocol that measures held-out success as a function of accumulated experience (an area-under-the-learning-curve metric), cross-embodiment transfer, and library-reuse rate under a common budget. Building such a benchmark is the natural companion to this survey. **(2) Adaptation as a first-class marketplace primitive.** The open problems of §4 suggest a research programme in which a downloaded skill is not a frozen artefact but a starting point that the §3.1 loop specialises to the local robot and environment—turning distribution (layer 2) and adaptation (layer 3) into a single pipeline. **(3) Bridging the RL and code flavours of skills.** §3.4 shows that skills are learned either as latent RL policies or as retrievable code; a unifying interface—code that wraps, calls, and refines learned policies, and learned policies distilled back into inspectable code—would let the self-improvement machinery of §3.1 operate over both. We see the combination of a standard self-improvement benchmark with an adaptation-centric marketplace as the most likely path from today’s static skill stores to genuinely capable, compounding robots.

7 CONCLUSION

Robot learning is organising itself around a single question: should competence be shipped as *weights* or as *skills*? This survey took that question as its axis. We mapped the field into six branches (§2), positioned end-to-end vision-language- action models as the weights-based foil (§3.2), and gave the code-as-policy camp its own deep-dive (§3.1) organised by *degree of self-improvement*: from one-shot program synthesis, through within-task repair, persistent skill memory, and population search, to the open-ended loop that combines all three. That top cell—feedback + memory + search—is the least populated and least surveyed region of the field, and it is where systems such as ASPIRE and ENPIRE now sit. We complemented this with a map of how skill *repertoires* are built (§3.4), showing that skills come in an RL flavour and a code flavour of which only the latter self-improves without gradients, and we connected the academic taxonomy to the nascent skill economy (§4), where commercial marketplaces already distribute skills but ship only static playback. The gap between that static distribution and genuine on-device adaptation is, we argue, the defining research opportunity for physical AI, and the self-improvement ladder of §3.1 is the road across it.

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